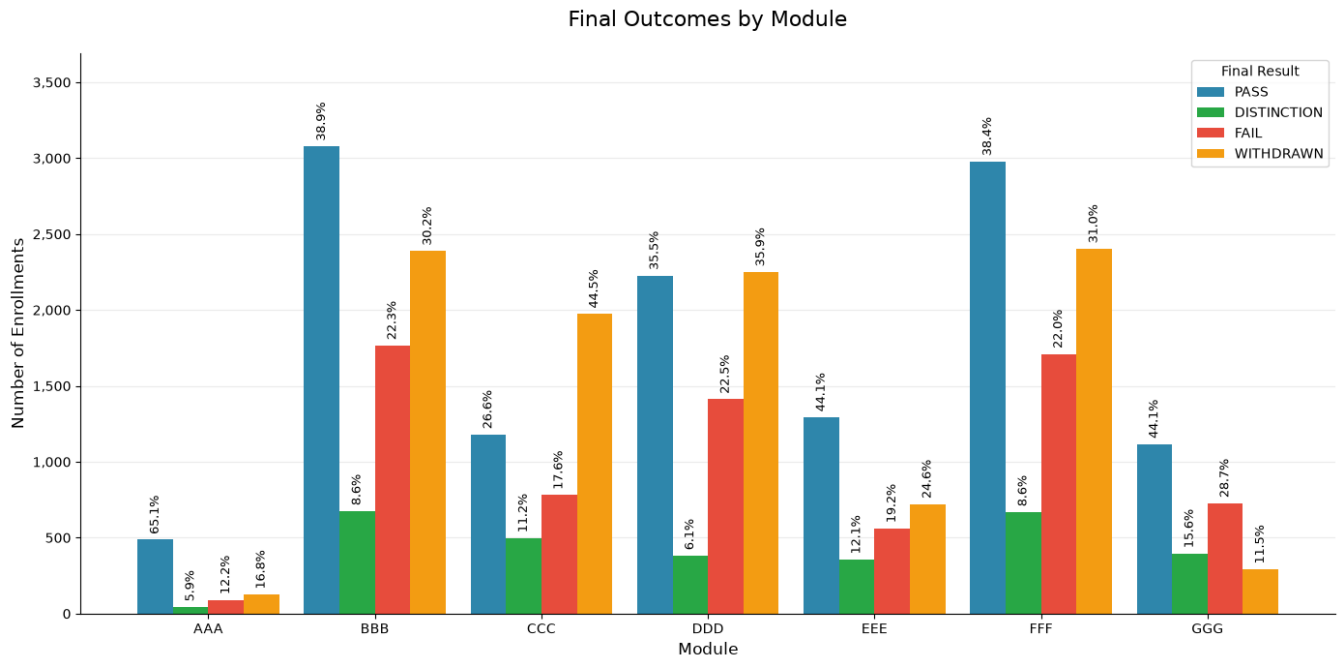
OULAD Charts and Interpretation

A companion to the executive summary. Historical associations do not establish causality. Success includes PASS and DISTINCTION.

Figure 1 Outcomes by module



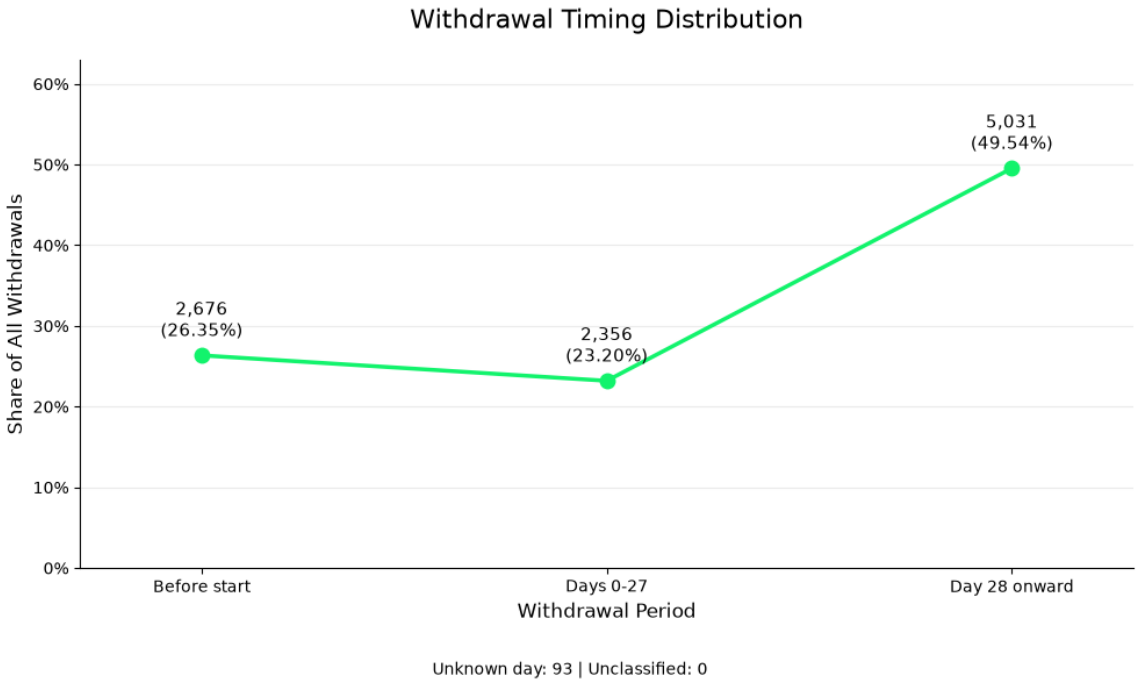
Axes and reading the chart. The horizontal axis lists modules AAA to GGG. The vertical axis shows enrollment counts. Colored bars distinguish PASS, DISTINCTION, FAIL and WITHDRAWN. Percentage labels use all enrollments within each module as the denominator; success combines PASS and DISTINCTION.

Main finding. CCC has the highest withdrawal rate at 44.54%. FFF has the largest withdrawal count, 2,403, compared with 1,975 in CCC. GGG has the highest failure rate, 28.73%, despite a withdrawal rate of only 11.52%.

Implications and limits. Prioritize withdrawal support in CCC and DDD, capacity for the case volumes in FFF and BBB, and academic support in GGG. A high bar may reflect a larger module rather than a higher risk; compare counts and percentages together.

Source: Final_OULAD_Analysis(3).ipynb, saved analysis outputs.

Figure 2 When withdrawals occurred



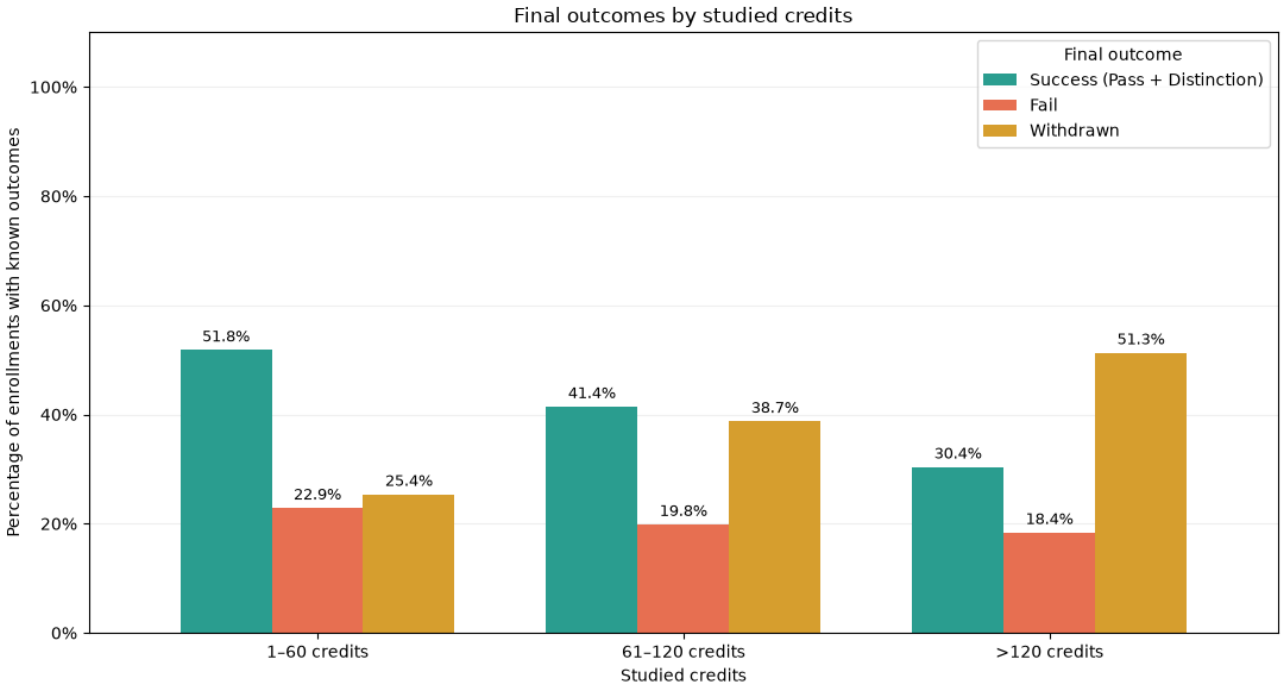
Axes and reading the chart. The horizontal axis groups withdrawals into before start, days 0–27 and day 28 onward. The vertical axis shows each period's share of all 10,156 withdrawals. Labels report counts and percentages; 93 withdrawals have an unknown day and are noted below the plot.

Main finding. Before-start withdrawals account for 26.35%, and days 0–27 account for another 23.20%. Together they represent 49.55% of withdrawals. A further 49.54% occurred from day 28 onward.

Implications and limits. Begin onboarding and access checks before day 0, provide early support, and continue monitoring later in the course. The periods have unequal lengths. The connecting line does not show a daily withdrawal hazard or a comparable rate per week.

Source: Final_OULAD_Analysis(3).ipynb, saved analysis outputs.

Figure 3 Outcomes by study load



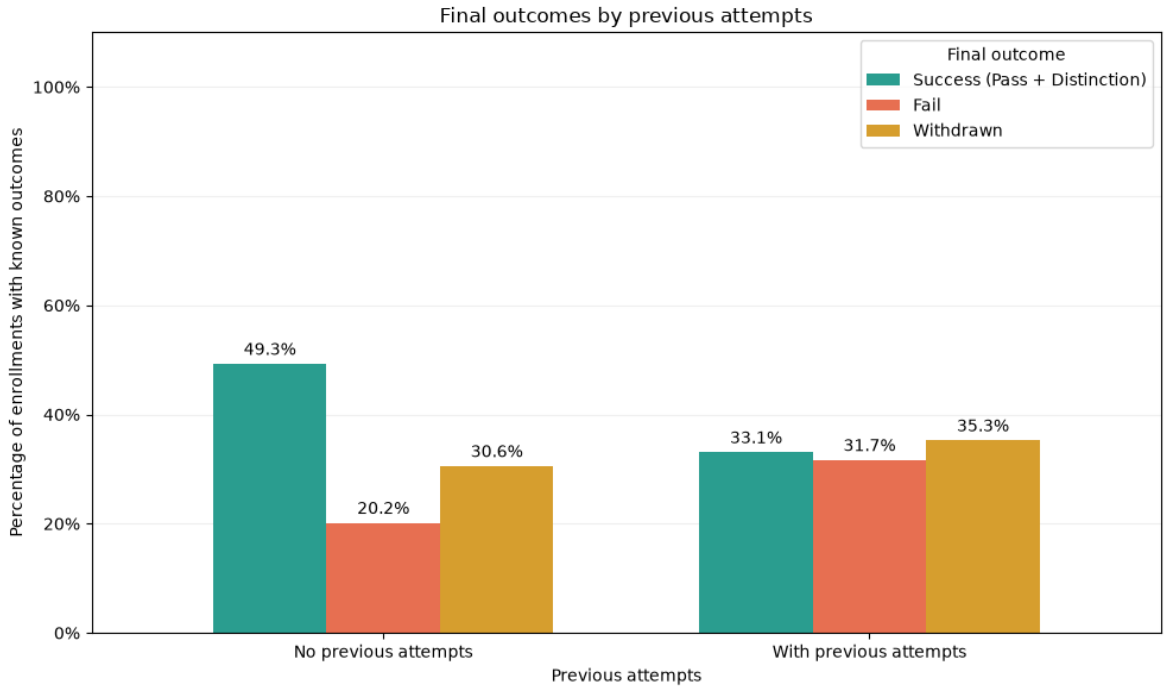
Axes and reading the chart. The horizontal axis groups StudiedCredits into 1–60, 61–120 and more than 120 credits. The vertical axis shows outcome percentages within each workload band. Colors identify success, failure and withdrawal. The bands contain 20,568, 9,769 and 2,256 enrollments, respectively.

Main finding. Success falls from 51.80% to 41.42% and then 30.36% as workload increases. Withdrawal rises from 25.35% to 38.75% and then 51.29%. Failure falls from 22.85% to 18.35%, alongside the larger withdrawal share.

Implications and limits. Review whether study load fits each student's available time, particularly above 120 credits. Lower failure in the highest-load group does not demonstrate better performance, because withdrawal is much higher. These descriptive differences do not establish that workload causes withdrawal.

Source: Final_OULAD_Analysis(3).ipynb, saved analysis outputs.

Figure 4 Outcomes by previous attempts



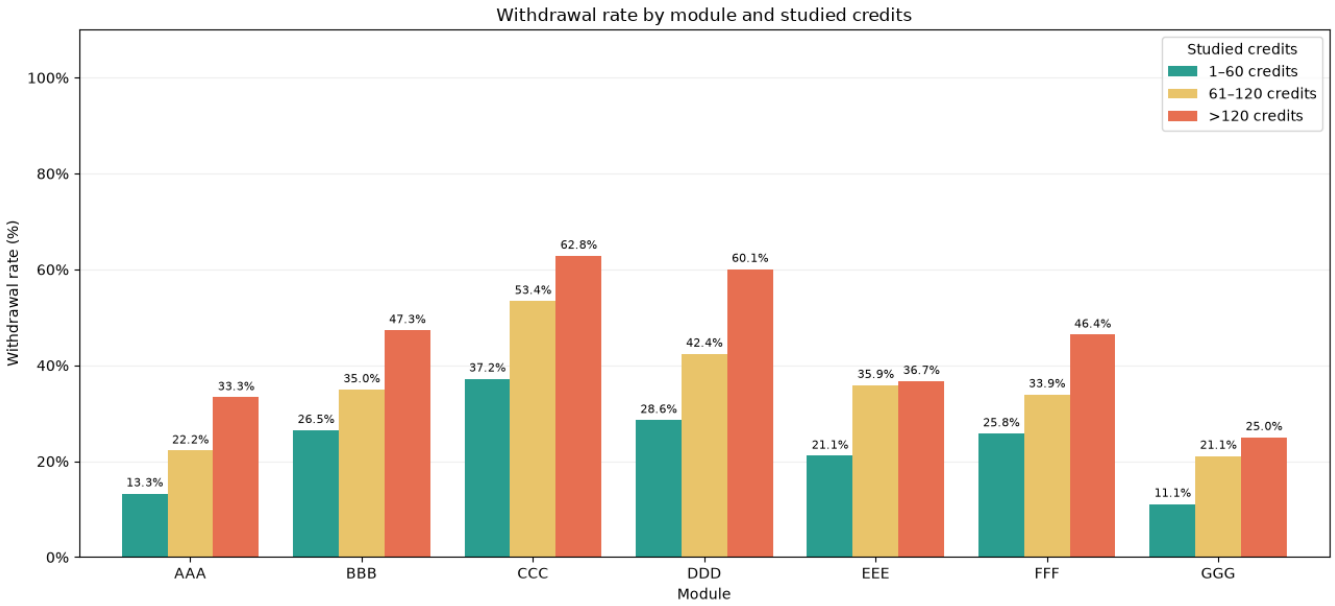
Axes and reading the chart. The horizontal axis distinguishes enrollments with no previous attempts from those with previous attempts. The vertical axis shows the percentage of enrollments in each group. Colors distinguish success, failure and withdrawal. Group sizes are 28,421 and 4,172 enrollments.

Main finding. Success is 49.28% without previous attempts and 33.08% with previous attempts. Failure rises from 20.16% to 31.66%, and withdrawal from 30.56% to 35.26%.

Implications and limits. Offer structured follow-up and academic preparation to students repeating a module. Previous attempts identify a group with weaker recorded outcomes; the comparison does not isolate their effect from workload, module or student characteristics.

Source: Final_OULAD_Analysis(3).ipynb, saved analysis outputs.

Figure 5 Withdrawal by study load within each module



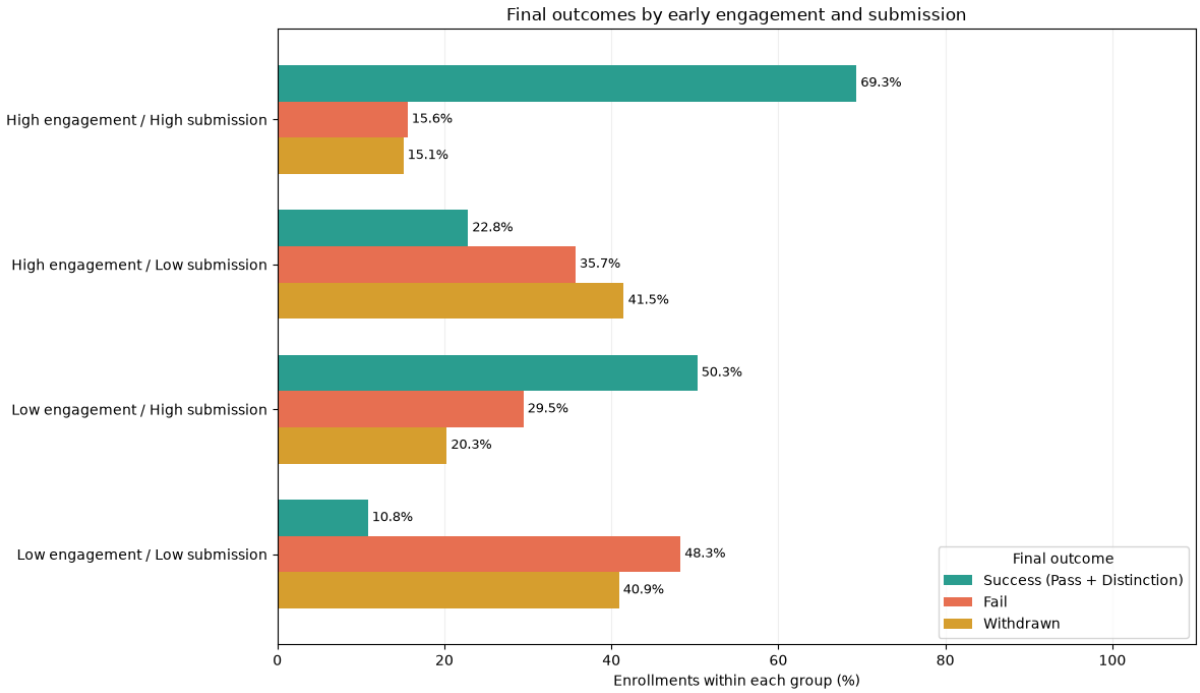
Axes and reading the chart. The horizontal axis lists the seven modules. The vertical axis shows withdrawal percentages within each module and workload band. Each module has three bars representing 1–60, 61–120 and more than 120 studied credits.

Main finding. Withdrawal rises across workload bands in all seven modules. In CCC it increases from 37.22% to 62.84%; in DDD, from 28.59% to 60.08%. DDD has the largest low-to-high workload gap, about 31.49 percentage points. CCC has the highest rate in every band.

Implications and limits. The overall workload pattern also appears within modules, supporting closer workload review in CCC and DDD. This does not adjust for presentation, previous attempts or student characteristics. Interpret small groups cautiously: GGG has only 12 enrollments above 120 credits.

Source: Final_OULAD_Analysis(3).ipynb, saved analysis outputs.

Figure 6 Outcomes by early engagement and submission



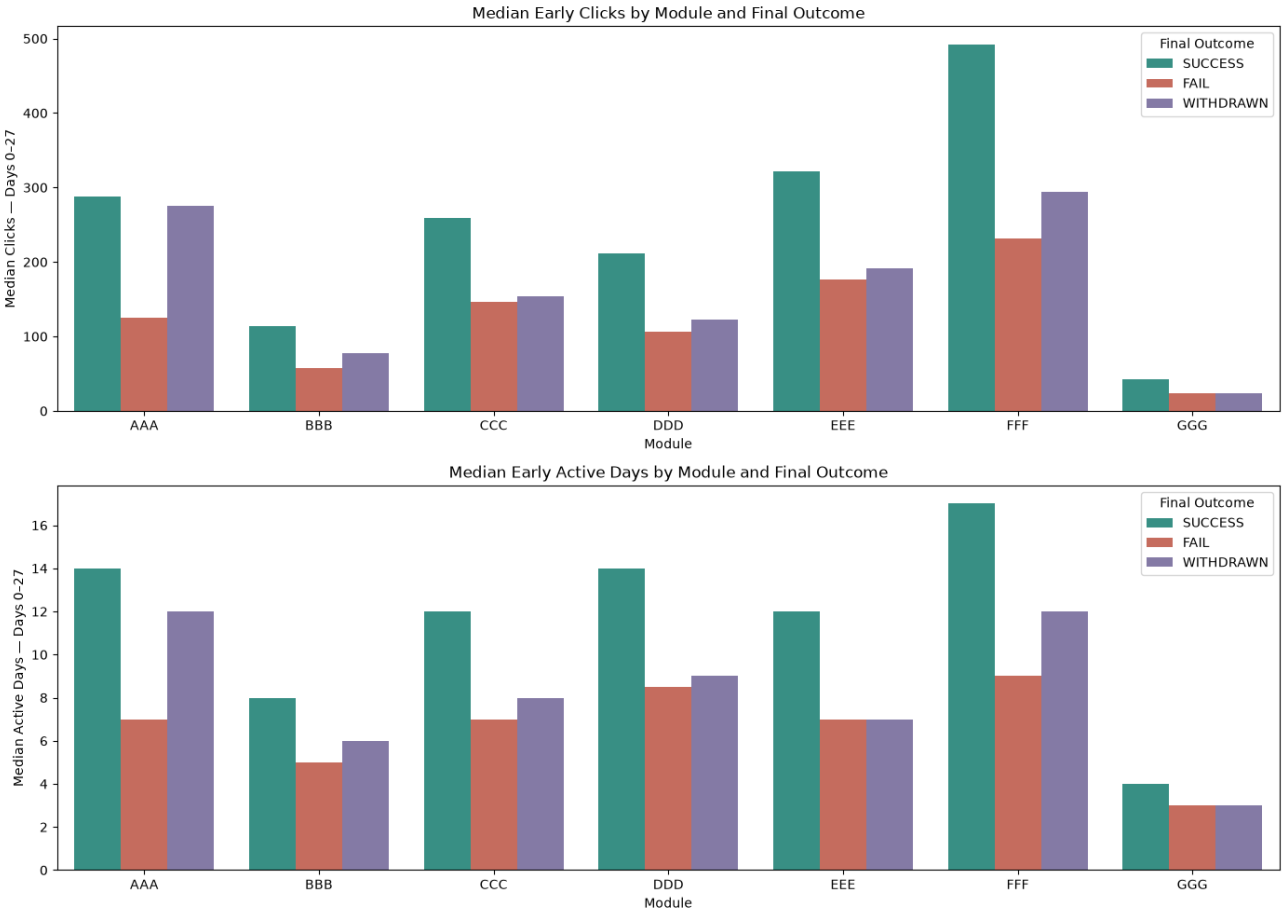
Axes and reading the chart. The vertical axis lists four combinations of high or low engagement and submission. The horizontal axis shows outcome percentages within each group. Colors identify success, failure and withdrawal. Measures use days 0–27: high engagement means positive clicks at or above the presentation median in this sample; high submission means at least 80% of assessments due in the window were submitted by day 27.

Main finding. Success is 69.35% when both indicators are high, compared with 22.82% when engagement is high but submission is low. With both indicators low, success is 10.83%, failure 48.27% and withdrawal 40.90%.

Implications and limits. Follow up assessment completion even when platform activity is high. The sample covers 21,533 enrollments in five modules and excludes EEE and GGG. The high-engagement / low-submission group has only 241 enrollments. Thresholds are exploratory, not validated alert rules.

Source: Final_OULAD_Analysis(3).ipynb, saved analysis outputs.

Figure 7 Early clicks and active days by module



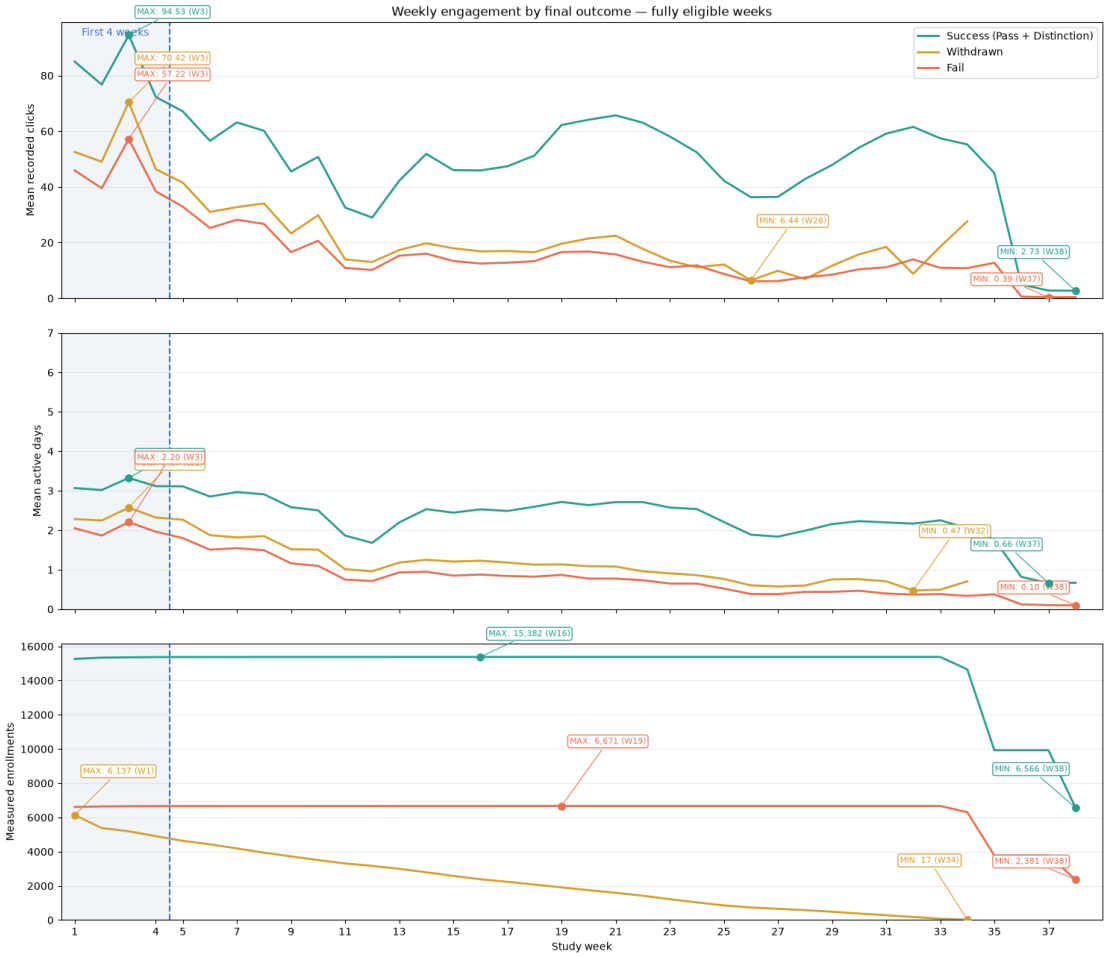
Axes and reading the chart. Both panels list modules on the horizontal axis. The upper vertical axis shows median clicks during days 0–27; the lower shows median active days in that window. Bar colors identify SUCCESS, FAIL and WITHDRAWN. SUCCESS combines PASS and DISTINCTION.

Main finding. The successful group has higher median clicks and active days than the failed and withdrawn groups in all seven modules. Absolute activity differs substantially between modules; for example, click levels are much higher in FFF than in GGG. In AAA, clicks among withdrawn and successful enrollments are relatively close.

Implications and limits. Use module-specific engagement patterns together with assessment information. The comparison covers 26,754 enrollments eligible for the full window and with interaction records; it excludes early withdrawals and 475 eligible enrollments without records. It cannot support a universal click threshold or a causal claim.

Source: Final_OULAD_Analysis(3).ipynb, saved analysis outputs.

Figure 8 Weekly engagement and the changing sample



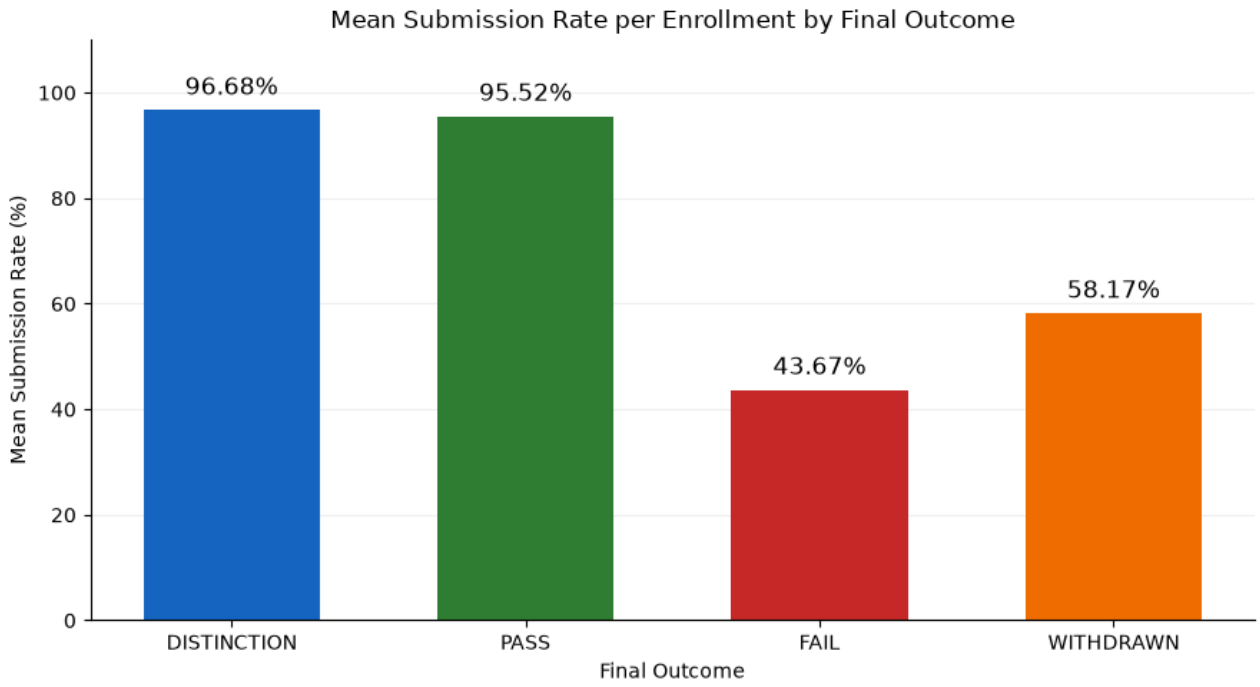
Axes and reading the chart. All three panels use study week on the horizontal axis. Vertical axes show mean recorded clicks, mean recorded active days, and measured enrollment counts, respectively. Colored lines identify SUCCESS, WITHDRAWN and FAIL. Means include full-week eligible enrollments with records at any time, including their zero-click weeks.

Main finding. The success group generally maintains higher weekly clicks and active days than the other groups. In week 3, mean clicks are 94.53 for success, 70.42 for withdrawal and 57.22 for failure. The measured withdrawal group shrinks over time as students leave.

Implications and limits. Monitor sustained changes in activity rather than one isolated week. The lines describe changing eligible groups, not fixed individual trajectories. Late-course fluctuations may reflect course calendars and sample composition. The withdrawal line ends when eligible weeks cease, rather than continuing at zero.

Source: Final_OULAD_Analysis(3).ipynb, saved analysis outputs.

Figure 9 Submission completion by final outcome



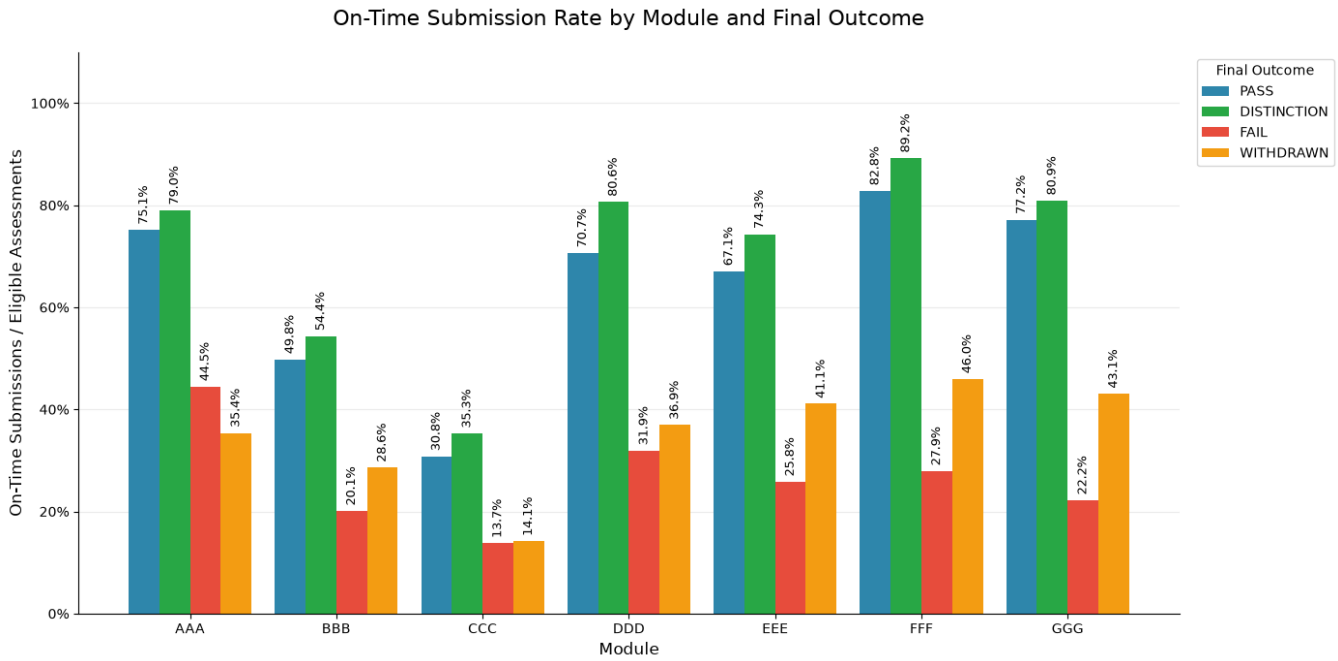
Axes and reading the chart. The horizontal axis lists DISTINCTION, PASS, FAIL and WITHDRAWN. The vertical axis shows the mean submission completion rate per enrollment. Each enrollment's submitted assessments are divided by its eligible current assessment opportunities, excluding banked assessments; these enrollment rates are then averaged within outcome groups.

Main finding. Mean completion is 96.68% for distinction, 95.52% for pass, 43.67% for failure and 58.17% for withdrawal. Failed enrollments show the lowest average completion, despite withdrawal being the larger issue overall.

Implications and limits. Track missing expected submissions as a potential support signal. These are enrollment averages, not one pooled assessment-level rate. For withdrawn enrollments, only assessments due while eligible are included. An absent submission record should be checked for completeness before it is treated as confirmed non-submission.

Source: Final_OULAD_Analysis(3).ipynb, saved analysis outputs.

Figure 10 On-time submission by module and outcome



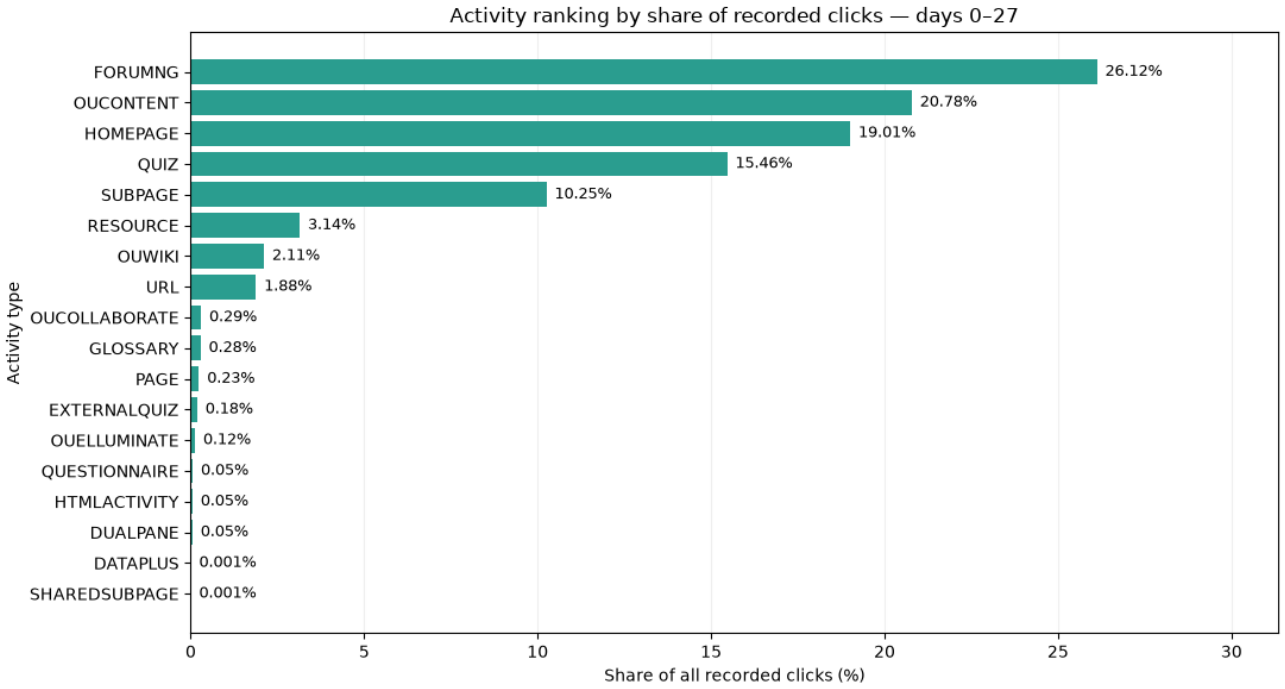
Axes and reading the chart. The horizontal axis lists modules AAA to GGG. The vertical axis shows on-time submissions as a percentage of eligible assessment opportunities within each module and outcome group. Colors distinguish PASS, DISTINCTION, FAIL and WITHDRAWN. The denominator includes opportunities without a submission record.

Main finding. In CCC, on-time submission is only 13.7% for failed enrollments and 14.1% for withdrawn enrollments. In FFF, it is 27.9% for failed enrollments versus 82.8% for pass and 89.2% for distinction. This highlights substantial variation across modules and outcomes.

Implications and limits. Check due-date definitions, exceptions and missing submissions before designing reminders or follow-up. A low percentage can reflect missing submissions as well as lateness. It is not the proportion on time among actual submissions, and differs from the enrollment-average completion measure in Figure 9.

Source: Final_OULAD_Analysis(3).ipynb, saved analysis outputs.

Figure 11 Resource types ranked by share of clicks



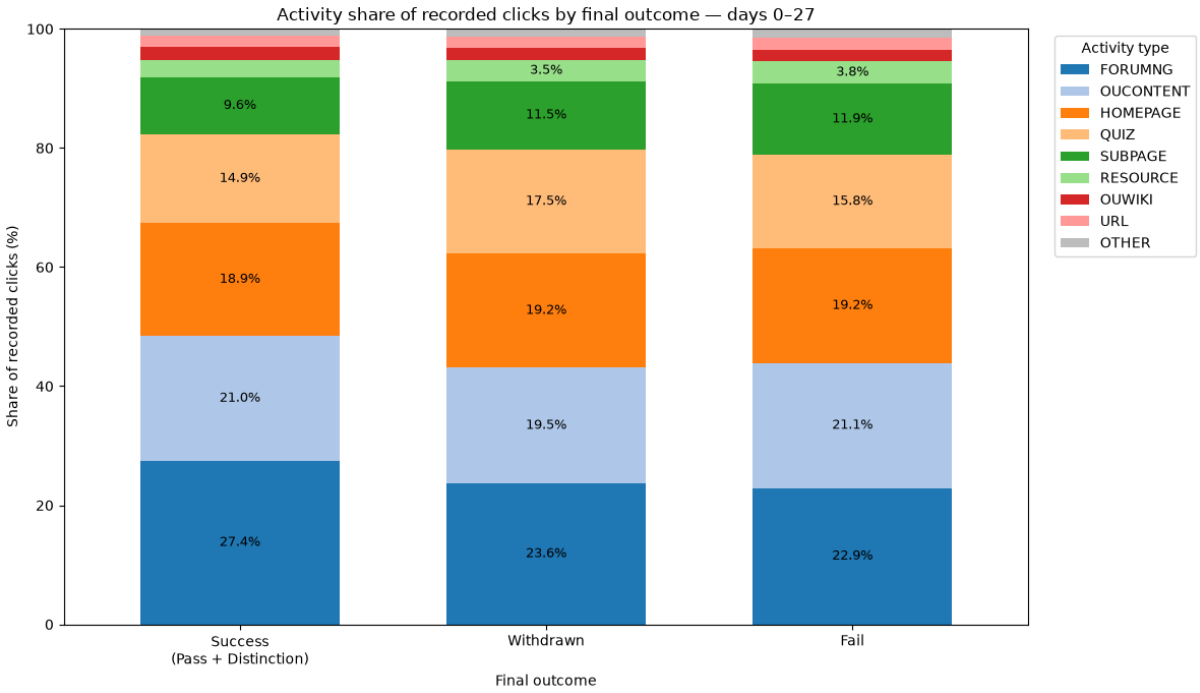
Axes and reading the chart. The vertical axis lists learning resource types, ranked by recorded click share. The horizontal axis shows each type's percentage of all clicks during days 0–27 in the analysis sample. The labels are click shares, not percentages of students using each resource.

Main finding. FORUMNG contributes 26.12% of clicks, followed by OUCONTENT at 20.78%, HOMEPAGE at 19.01% and QUIZ at 15.46%. Together these four types account for approximately 81.37% of recorded clicks, allowing for rounding.

Implications and limits. Prioritize usability and support checks for these heavily used resources. The sample contains 26,754 full-window eligible enrollments with records. Click totals reflect resource design and use, not necessarily learning quality. Low use does not by itself show that a resource is ineffective or unnecessary.

Source: Final_OULAD_Analysis(3).ipynb, saved analysis outputs.

Figure 12 Resource click composition within outcome groups



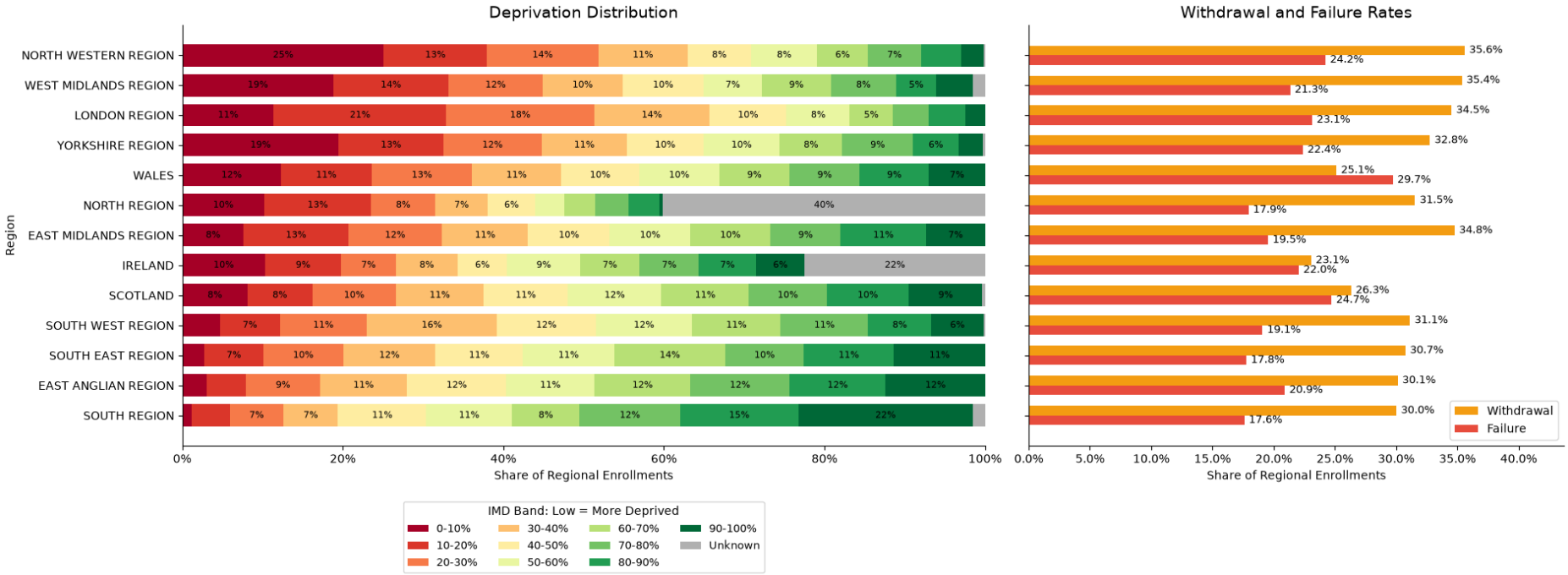
Axes and reading the chart. The horizontal axis lists Success, Withdrawn and Fail. The vertical axis runs from 0% to 100%. Each stacked column shows how all clicks within that outcome group are distributed across resource types; colors identify the types, with less common resources combined as OTHER.

Main finding. FORUMNG represents 27.44% of successful enrollments' clicks, compared with 23.64% for withdrawn and 22.87% for failed enrollments. QUIZ represents 14.93%, 17.49% and 15.81%, respectively. The overall composition is broadly similar, with differences in some resource shares.

Implications and limits. Test whether assessment-related discussion and easier content access improve completion, rather than assuming a higher resource share causes success. A 100% column shows composition, not total clicks or users. Comparisons may also reflect differences in available resources and module mix; the sample is restricted to eligible enrollments in days 0–27.

Source: Final_OULAD_Analysis(3).ipynb, saved analysis outputs.

Figure 13 Deprivation and outcomes by region



Axes and reading the chart. Regions form the shared vertical axis. The left horizontal axis shows the percentage of enrollments in each deprivation band within each region; each stacked bar totals 100%. Lower IMD bands mean greater deprivation, and gray indicates an unknown band. On the right, the horizontal axis shows withdrawal and failure percentages within the same region. Orange represents withdrawal; red represents failure. Rows align across panels, and each region's enrollments are the denominator.

Main finding. North Western Region has 37.9% of enrollments in the most deprived 0–20% bands, with 35.6% withdrawal and 24.2% failure. South Region has 5.9% in those bands, with 30.0% withdrawal and 17.6% failure. The pattern is not uniform: Wales has the highest regional failure rate, 29.7%, despite a lower concentration in the most deprived bands than North Western Region.

Implications and limits. Review regional access to academic support, but do not attribute the differences to deprivation alone. IMD band is missing for 40.1% of North Region enrollments and 22.5% in Ireland, limiting comparisons. These are regional summaries of the project's enrollments, not population estimates or individual-level effects. Compare deprivation bands within regions and account for module and presentation differences before drawing stronger conclusions.

Source: Final_OULAD_Analysis(3).ipynb, saved analysis outputs.